Software Requirements Specification

for

Veridex

A Rule-Based Public Information Reliability & Verification System

Version 1.0 approved

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Revision History

Name	Date	Reason for Changes	Version
Initial Draft	4/01/2026	Initial SRS creation	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for **Veridex**, Version 1.0, a rule-based public information reliability and verification system. This document describes the scope, functionality, and constraints of the software, including claim submission, source validation, rule-based reliability evaluation, conflict detection, freshness checks, and anonymous community interaction features, and serves as a reference for design, development, testing, and academic evaluation.

1.2 Document Conventions

This document follows the standard Software Requirements Specification (SRS) format proposed by Karl E. Wiegers. Consistent section numbering, headings, and formatting are used to improve clarity and ease of reference. Functional requirements are expressed using the keyword **“shall”** to indicate mandatory system behaviour. Each requirement is written in clear, unambiguous language and is uniquely identifiable. Higher-level requirements are considered applicable to their related detailed requirements unless explicitly stated otherwise. No special typographical symbols or color coding are used, ensuring the document remains simple, readable, and suitable for academic evaluation.

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification is intended for faculty evaluators, project guides, student developers, testers, and reviewers involved in the design and assessment of the system. In addition, this document is structured to support future intellectual property review by clearly defining the system scope, functionality, and technical boundaries. Faculty members and evaluators may use this document to assess the problem definition, innovation, and feasibility of the proposed system. Developers are advised to read the document sequentially, starting with the introduction and overall description to understand the system context, followed by system features and non-functional requirements for implementation details. Reviewers and testers may focus on the functional and interface requirements to validate system behaviour and consistency. The document is organized to provide a clear and logical flow from high-level concepts to detailed requirements.

1.4 Product Scope

Veridex is a rule-based software system designed to evaluate the reliability of publicly shared information by applying deterministic verification logic rather than subjective judgment or artificial intelligence. The primary objective of the system is to assist users in identifying whether public information is reliable, outdated, or conflicting based on source validation, temporal relevance, and consistency checks. The system aims to improve information awareness, reduce the impact of misinformation, and support informed decision-making at a large scale. From a broader perspective, the project aligns with societal and institutional goals of information transparency, digital trust, and responsible information dissemination, while being structured to support future intellectual property evaluation and extension.

1.5 References

- Karl E. Wiegers, *Software Requirements*, Microsoft Press, 2nd Edition, 1999.
- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
- Course material and lecture notes for Software Engineering, School of Information Technology & Engineering, VIT Vellore.
- Course material and laboratory manuals for Web Technologies, School of Information Technology & Engineering, VIT Vellore.
- Internal project proposal and vision notes prepared by the project team for Veridex (unpublished).

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2. Overall Description

2.1 Product Perspective

Veridex is a new, standalone software system developed to address the problem of assessing the reliability of publicly shared information. It is not a replacement for any existing system, nor is it a direct extension of a current product family. The system is designed as an independent web-based application that integrates a rule-based verification engine, data storage, and user interaction components within a client–server architecture. While Veridex operates as a self-contained product, it is structured to interface with external information sources such as official websites and public media links through user-provided references. The modular design of the system allows future integration with larger information platforms or institutional systems without modifying the core verification logic.

2.2 Product Functions

The major functions of the Veridex system include the following:

- Allow users to submit public information claims along with supporting sources
- Classify and manage different types of information sources
- Evaluate the reliability of claims using predefined, rule-based verification logic
- Detect outdated information and conflicting claims within the system
- Generate and display confidence indicators and verification status for each claim
- Enable anonymous community interaction through discussions and polls to capture real-world persistence and context
- Maintain a verification history and audit trail for claims and updates
- Support moderation and administrative control for maintaining system integrity

These functions collectively support the objective of providing a structured and transparent mechanism for assessing public information reliability.

2.3 User Classes and Characteristics

Actor	Description	Key Responsibilities
General User	Primary system user	Submit claims, view reliability status, participate anonymously
Anonymous Community Participant	Non-identified user	Participate in polls and discussions related to claim persistence
Contributor	Privileged user	Add or update sources, provide clarifications
Moderator	Supervisory user	Monitor claims, manage misuse, oversee discussions
Administrator	System manager	Manage rules, roles, and system configuration

Actor	Submit Claim	View Status	Chat/Poll	Moderate	Configure
General User	✓	✓	✓	✕	✕
Anonymous User	✕	✓	✓	✕	✕
Contributor	✓	✓	✓	✕	✕
Moderator	✕	✓	✓	✓	✕
Administrator	✕	✕	✕	✓	✓

2.4 Operating Environment

The Veridex system is designed to operate as a web-based application accessible through standard web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox. The software runs on client devices including desktop and laptop computers with commonly used operating systems such as Windows, Linux, or macOS. The backend services are hosted on a server environment supporting Node.js and MongoDB, and the system communicates over standard HTTP/HTTPS

protocols. The application is intended to coexist with other web-based applications without requiring any specialized hardware or platform-specific dependencies.

2.5 Design and Implementation Constraints

The design and implementation of the Veridex system are subject to several constraints. The system must employ a deterministic, rule-based verification approach and shall not rely on artificial intelligence or machine learning techniques. The application is required to be developed using web technologies, with a client–server architecture and a NoSQL database for data storage. Standard web communication protocols such as HTTP and HTTPS must be used to ensure compatibility and security. User privacy must be preserved through anonymous interaction mechanisms, and role-based access control must be enforced. Additionally, the system design must follow secure coding practices and maintain modularity to support future extensions without altering the core verification logic.

2.6 User Documentation

The Veridex system will be accompanied by user documentation to assist different user classes in understanding and operating the application. This documentation will include a basic user guide explaining claim submission, viewing reliability indicators, and participating in anonymous discussions and polls. Online help content will be integrated within the application to provide contextual assistance for key features. Administrative and moderation guidelines will also be provided to describe role-specific responsibilities. All documentation will be delivered in digital format and written in clear, user-friendly language suitable for academic and demonstration purposes.

2.7 Assumptions and Dependencies

The development of the Veridex system assumes that users will provide publicly accessible information and valid external sources while submitting claims. It is assumed that reliable internet connectivity will be available for both users and the hosting environment. The system depends on standard web technologies and third-party libraries used within the frontend and backend frameworks, as well as the availability of a NoSQL database for data storage. The effectiveness of certain system functions, such as source validation and freshness checks, depends on the accuracy and accessibility of referenced external websites. Changes in development tools, browser compatibility, or hosting environments may require corresponding adjustments to the system implementation.

3. External Interface Requirements

3.1 User Interfaces

The Veridex system provides a web-based user interface designed for ease of use and clarity across all user classes. The interface includes dedicated screens for claim submission, claim listing, detailed claim views, reliability status display, and anonymous community interaction through discussions and polls. Consistent layout, navigation menus, and standard action buttons such as submit, view, and help are used throughout the application. Error messages and system notifications are displayed in a clear and informative manner to guide user actions. The user interface is designed to be responsive and accessible through common web browsers. Detailed interface layouts and visual designs are maintained separately in a user interface specification document.

3.2 Hardware Interfaces

The Veridex system does not require any specialized hardware interfaces. The software is designed to operate on standard computing devices such as desktop and laptop computers, and on mobile devices when accessed through a web browser. Interaction between the software and hardware is limited to standard input and output operations, including keyboard, mouse, and touch-based inputs. All data processing is handled at the application and server level, with communication occurring over standard network interfaces using HTTP and HTTPS protocols.

3.3 Software Interfaces

The Veridex system interfaces with several software components to support its operation. The frontend of the system communicates with the backend through RESTful web services using standard HTTP and HTTPS protocols. The backend interacts with a NoSQL database to store and retrieve data related to claims, sources, verification status, and community interactions. Standard web browsers act as the client interface for rendering the user interface and handling user input. Third-party libraries and frameworks are used to support frontend rendering, backend request handling, and database connectivity. Data exchanged between software components includes claim details, source metadata, reliability indicators, poll responses, and discussion entries, all of which are transmitted in structured formats. Detailed API specifications and data models are maintained separately as part of the system design documentation.

3.4 Communications Interfaces

The Veridex system relies on standard web-based communication mechanisms for user interaction and backend processing. All communication between the client and server is conducted through HTTP and HTTPS protocols using a web browser as the primary access interface. Data exchanged between system components, including claim submissions, source details, reliability indicators, poll responses, and discussion content, is transmitted in structured formats such as JSON. Secure communication is ensured through HTTPS encryption to protect data integrity and user privacy. No email-based or file transfer communication is required for core system functionality, and network communication is designed to support reliable, asynchronous request–response interactions.

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4. System Features

This section describes the major functional features provided by the Veridex system. The system features are organized based on core system services and verification workflows to clearly represent system behaviour and responsibilities.

4.1 Claim Submission and Source Management

4.1.1 Description and Priority

This feature enables users to submit public information claims along with supporting sources for verification. It serves as the primary input mechanism for the system and is essential for all subsequent processing.
Priority: High

4.1.2 Stimulus/Response Sequences

- User submits a public information claim with one or more sources.
- The system validates the input format and source entries.
- The system stores the claim and associated source data in the database.
- The system confirms successful submission or displays appropriate error messages.

4.1.3 Functional Requirements

- REQ-1: The system shall allow users to submit public information claims.
- REQ-2: The system shall allow users to attach one or more supporting sources to a claim.

- REQ-3: The system shall validate claim and source input before storage.
- REQ-4: The system shall store claim submission timestamps for verification purposes.

4.2 Rule-Based Reliability Evaluation

4.2.1 Description and Priority

This feature evaluates the reliability of submitted claims using predefined, deterministic rules based on source diversity, source type, temporal freshness, and conflict detection. This feature represents the core verification logic of the system.
Priority: High

4.2.2 Stimulus/Response Sequences

- A claim is submitted or updated.
- The system applies rule-based verification logic.
- The system computes a reliability status and confidence indicator.
- The system stores and displays the evaluation results.

4.2.3 Functional Requirements

- REQ-5: The system shall evaluate claims using predefined rule-based logic.
- REQ-6: The system shall assess source diversity and source type during evaluation.
- REQ-7: The system shall detect outdated information based on time thresholds.
- REQ-8: The system shall identify conflicting claims within the system.
- REQ-9: The system shall generate a reliability status without determining absolute truth.

4.3 Anonymous Community Interaction

4.3.1 Description and Priority

This feature enables anonymous, claim-specific community interaction through discussions and polls to capture real-world context and issue persistence. This feature supports verification workflows without influencing reliability decisions directly.
Priority: Medium

4.3.2 Stimulus/Response Sequences

- User accesses a claim's interaction section.
- User participates anonymously in discussions or polls.
- The system records responses and displays aggregated context indicators.

4.3.3 Functional Requirements

- REQ-10: The system shall allow anonymous discussion under individual claims.
- REQ-11: The system shall provide polling mechanisms to capture issue persistence.
- REQ-12: The system shall prevent community interactions from directly modifying reliability scores.

4.4 Moderation and Administration

4.4.1 Description and Priority

This feature provides moderation and administrative controls to manage claims, interactions, and system configuration. It ensures system integrity, security, and controlled operation.
Priority: Medium

4.4.2 Stimulus/Response Sequences

- Moderator reviews claims or community activity.
- Administrator updates system rules or user roles.
- The system applies changes and logs actions for auditing.

4.4.3 Functional Requirements

- REQ-13: The system shall support role-based access control.
- REQ-14: The system shall allow moderators to manage misuse and violations.
- REQ-15: The system shall maintain an audit trail of administrative actions.

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Veridex system is expected to support multiple concurrent users with acceptable response times under normal operating conditions. User-facing actions such as claim submission, claim retrieval, and viewing reliability indicators should be processed within a reasonable time to ensure smooth interaction. Backend rule-based evaluation should execute efficiently upon claim submission or update without noticeable delay to the user. The system should be capable of handling increased data volume over time while maintaining consistent performance. These performance requirements ensure usability and scalability while allowing developers to make appropriate design and optimization decisions.

5.2 Safety Requirements

The Veridex system shall be designed to minimize the risk of misuse, misinformation amplification, or unintended harm resulting from incorrect interpretation of information. The system shall clearly indicate that reliability indicators are not judgments of absolute truth and should not be used as official or legal validation. Safeguards such as moderation controls and misuse reporting mechanisms shall be provided to prevent abusive or harmful content. The system shall avoid actions that could mislead users into assuming authoritative endorsement of information.

5.3 Security Requirements

The Veridex system shall enforce role-based access control to restrict sensitive operations to authorized users. User interactions shall be protected through secure authentication mechanisms for privileged roles such as contributors, moderators, and administrators. Anonymous participation shall be supported without storing personally identifiable information. All data transmission between system components shall be secured using encryption through HTTPS. The system shall protect stored data from unauthorized access and comply with standard data protection and privacy practices applicable to web-based applications.

5.4 Software Quality Attributes

The system shall emphasize reliability, maintainability, and usability as primary quality attributes. Veridex shall be designed to be modular, allowing future

enhancements without affecting core verification logic. The application shall be portable across major web browsers and operating environments. High usability shall be prioritized to ensure that users with minimal technical knowledge can interact with the system effectively. The system shall also be testable, enabling verification of individual rule components and overall system behaviour.

5.5 Business Rules

The Veridex system shall operate under predefined business rules that govern user roles and system behaviour. Reliability evaluations shall be determined exclusively by rule-based logic and shall not be influenced by user opinions, poll outcomes, or discussion activity. Anonymous community interactions shall be claim-specific and non-personal. Only authorized roles shall be permitted to modify verification rules or system configuration. These rules ensure consistent, transparent, and controlled operation of the system.

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6. Other Requirements

The Veridex system shall store structured data related to claims, sources, reliability evaluations, and community interactions in a database designed to support efficient retrieval and long-term scalability. The system shall maintain verification history and audit records for traceability purposes. Legal and ethical considerations require the system to clearly state that it does not provide official, legal, or authoritative validation of information. The software shall be developed in a modular manner to support reuse of core components, particularly the rule-based verification engine, in future systems or extensions. Internationalization and multilingual support may be considered as future enhancements but are not mandatory for the current scope.

Appendix A: Glossary

- Claim: A piece of publicly shared information submitted to the system for reliability evaluation.
- Source: Any reference or link provided in support of a claim, such as official documents, media reports, or public notices.

- Reliability Indicator: A system-generated status that reflects the assessed reliability of a claim based on predefined rules.
- Rule-Based Verification: A deterministic method of evaluation that applies fixed logical rules rather than artificial intelligence or machine learning.
- Conflict Detection: The process of identifying inconsistencies between multiple claims or sources related to the same topic.
- Temporal Freshness: The relevance of information based on the time of publication or last verification.
- Anonymous Interaction: User participation in discussions or polls without disclosure of personal identity.
- Moderator: A privileged user responsible for overseeing system usage and preventing misuse.
- Administrator: A user with full system access responsible for configuration and maintenance.

Appendix B: Analysis Models

The analysis models for the Veridex system include high-level representations used to understand system behavior and structure. These models may include use case diagrams to represent user interactions, data flow diagrams to illustrate information movement within the system, and architectural diagrams to show the relationship between system components. These models are maintained separately as part of the system design documentation and support both development and evaluation activities.

Appendix C: To Be Determined List

1. Final threshold values for rule-based reliability classification.
2. Exact user interface layout and visual design specifications.
3. Detailed moderation policies and escalation rules.
4. Final deployment and hosting configuration details.